Solutions to JEE Main – 2 |JEE - 2022

PHYSICS

SECTION-1

1.(B) $a = 4x$

$$\frac{vdv}{dx} = 4x$$

$$\int_0^v vdv = \int_0^x 4xdx$$

$$\frac{v^2}{2} = \frac{4x^2}{2}$$

$$v^2 = 4x^2$$

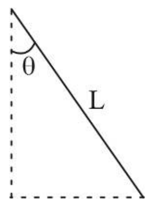
$$v^2 = 400$$

$$v = 20\ m/s$$

2.(C) $R = L\sin\theta$

(1) $T\sin\theta = mv^2/R$

(2) $T\cos\theta = mg$

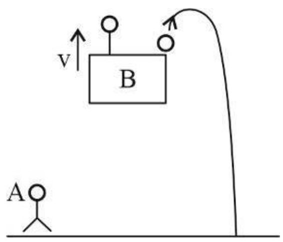


$$\tan\theta = \frac{mv^2}{mgR} \Rightarrow \tan\theta = \frac{v^2}{(L\sin\theta)g}$$

$$v = \sqrt{Lg\tan 30^\circ \sin 30^\circ} = \left(\frac{Lg}{2\sqrt{3}}\right)^{1/2}$$

3.(A) Force required to stop the liquid $= \rho A V^2$.

4.(D)

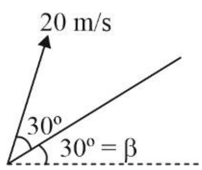


At the time of release both ball and person B has same speed $v_{rel} = 0$. According to B ball has free fall. For A ball will move up for some time.

5.(D) Slope of $x - t$ graph is velocity
Slope at $t = 8$ is zero hence velocity = 0

6.(B) $v_y = 20\sin 30^\circ = 10\ m/s$

$$a_y = g\cos\beta = \frac{g\sqrt{3}}{2}$$



$$\text{Time of flight} = 2\frac{v_y}{a_y} = 2 \times \frac{10}{\left(\frac{10\sqrt{3}}{2}\right)} = \frac{4}{\sqrt{3}}$$

7.(A) $H = \dfrac{v^2 \sin^2 \theta}{2g}, \; R = \dfrac{v^2}{g} \sin 2\theta$

On solving

$\tan \theta = \dfrac{4}{3} \qquad \text{so } \sin \theta = \dfrac{4}{5}$

$4 = \dfrac{v^2}{2g}\dfrac{16}{25} \Rightarrow \sqrt{\dfrac{200g}{16}} = v$

$v = 5\sqrt{\dfrac{g}{2}}$

8.(C) $x = 6t\,, \qquad x = v\cos\theta t$

$y = 8t - 5t^2 \qquad y = v\sin\theta \cdot t - \dfrac{1}{2}gt^2$

On comparing

$8 = v\sin\theta$

$6 = v\cos\theta$

$\tan\theta = \dfrac{4}{3}$

So, $\sin\theta = \dfrac{4}{5} \qquad v = 10 \; m/s$

9.(B) $H = \dfrac{v^2}{2g}\sin^2\theta$

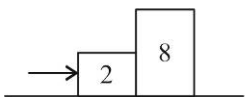
For same range $\theta_1 + \theta_2 = 90^\circ = \dfrac{\pi}{2}$

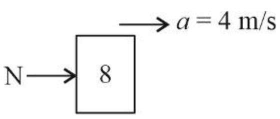
for $\theta_1,\; h_1 = \dfrac{v^2}{2g} \times \dfrac{3}{4}$

for $\theta_2,\; h_2 = \dfrac{v^2}{2g} \times \dfrac{1}{4} = \dfrac{h_1}{3}$

10.(B) Common acceleration of system $= \dfrac{Force\ external}{Total\ mass}$

$\dfrac{40}{10} = 4\; m/s^2$

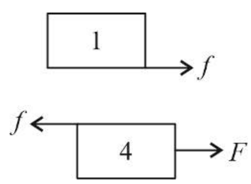




So $N = 8 \times 4 = 32N$

11.(B) Max acceleration of 1 kg $\Rightarrow \; f = ma$

$f = \mu N = ma \qquad a = \mu g = 2m/s^2$



So to move together whole system should move together with $a = 2m/s^2$

$F = 5 \times 2 = 10N$

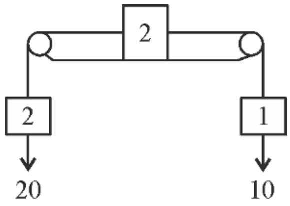
12.(C) Net force on system = 10 N

Total mass = 5 kg

Acceleration of system $= \frac{10}{5} = 2$

$T - 10 = 1 \times 2$

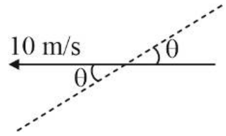
$T = 12N$



13.(B) Velocity along the rope is $10\cos\theta$

All velocities along rope are equal

$10\cos\theta = 5, \quad \theta = \frac{\pi}{3} = 60^o$



14.(C) To avoid slipping

$a = g\tan\theta$

$10 = 10\tan\theta$

$\theta = 45^o$

15.(D) $a = g\sin\theta - \mu_k\, g\cos\theta = \frac{g}{\sqrt{2}} - \frac{1}{3}\frac{g}{\sqrt{2}} = \frac{g}{\sqrt{2}}\left(\frac{2}{3}\right)$

distance $= 10\sqrt{2}m$

$10\sqrt{2} = \frac{1}{2}\frac{g}{\sqrt{2}}\left(\frac{2}{3}\right)t^2$

$t^2 = 6$

$t = \sqrt{6}\,\text{sec}$

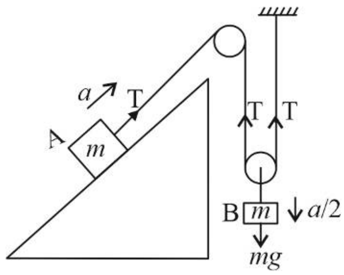
16.(D) By constraint acceleration will not be same

(1) $mg - 2T = \frac{ma}{2}$

(2) $T - mg\sin 30^o = ma$

On solving $a = 0$

So, acceleration of block B is $a = 0$



17.(D) Area under $v - t$ is displacement

$A = \frac{1}{2} \times 100 \times 500 = 25\ km$

18.(A) $u = V,\ V_f = -4V,\ a = -g,\ h = ?$

$(4V)^2 = (V)^2 + 2(-g)h$

$h = -\frac{15V^2}{2g}$ ('–' ve sign shows that displacement is downward)

19.(A) $v = 2t$

$a_t = 2$

$v(2) = 4m/s$

$a_v = \frac{v^2}{R} = 16$

$a_{total} = (16^2 + 2^2)^{1/2} = (260)^{1/2} = (4 \times 65)^{1/2} = 2\sqrt{65}\ m/s^2$

20.(C) $v = 4t^2$

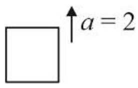
$p = mv = 8t^2$

$F = \frac{dp}{dt} = 16t \qquad F(t=4) = 64N$

SECTION-2

1.(4) $V_{rel} = 24\ m/s$

$a_{rel} = (10+2) = 12\ m/s^2$

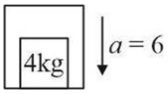


$t = \left(\frac{2v_{rel}}{a_{rel}}\right) = 2 \times 2 = 4$ sec

2.(8) Area of F-t graph = change is momentum = 12 kg m/s

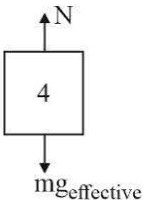
$m(V_f - V_i) = 12 \Rightarrow V_f = 8m/s$

3.(16)



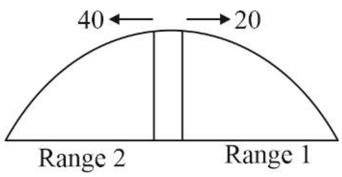
$g_{(effective)} = 10 - 6 = 4$

So, FBD of block in lift's frame



$\Rightarrow \quad N = 4 \times 4 = 16$

4.(24)



Time of flight $= \sqrt{\frac{2H}{g}} = 4$ as $v_y = 0$

Range 1 $= 20 \times 4 = 80$

Range 2 $= 40 \times 4 = 160$

Total distance between than $= 240\ m$

$= 10x = 240$

$x = 24$

5.(10) $mg = N = 100g$

$\mu N = \frac{mV_{max}^2}{R}$

$\mu mg = \frac{mV_{max}^2}{R}$

$5 = \frac{V_{max}^2}{20}$

$V_{max} = 10m/s$

CHEMISTRY

SECTION-1

1.(C) (I) (Fluorine), (II) (Oxygen), (III) (Chlorine), (IV) (Phosphorus)
Chlorine has highest value of 1[st] Electron gain enthalpy (magnitude)

2.(B) $\underset{0}{H} - \underset{0}{O} - \underset{0}{N} = \underset{0}{O}$

3.(B) $\mu_{theo} = 4.8 \times 10^{-10} \times 1.6 \times 10^{-8} = 7.68 \times 10^{-18} \text{esu.cm} = 7.68\text{ D}$

% ionic character $= \dfrac{3.2}{7.68} \times 100 = 41.6\%$

4.(D) $\text{I.E.}_4 \text{ of Be} = \left(\dfrac{Z^2}{n^2} \times 13.6 \times 1.6 \times 10^{-19}\right)\text{J} = 16 \times 13.6 \times 1.6 \times 10^{-19} = 348.16 \times 10^{-19}$

$= Y \times 10^{-19}\text{J} \qquad \therefore \qquad Y = 348.16$

5.(D) w = work function of potassium

$E_1 = \dfrac{hc}{\lambda_1} = w + (0.32 \times 10^{-19})$

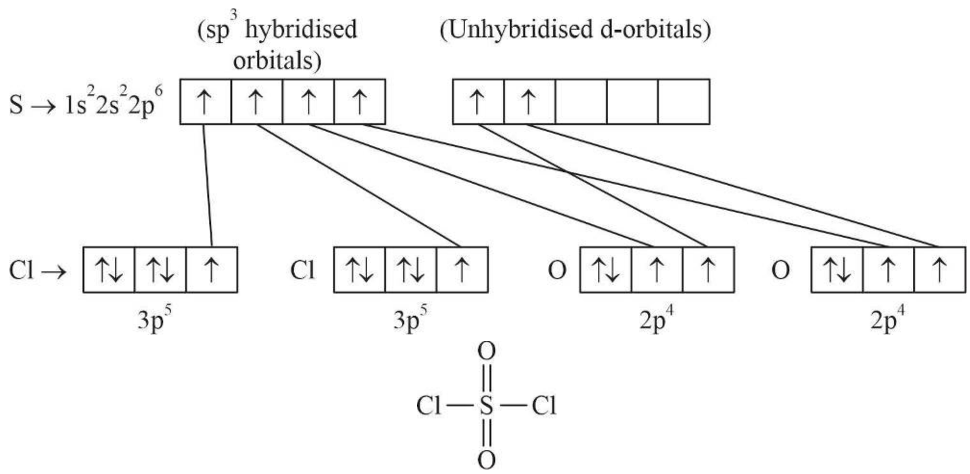
$E_2 = \dfrac{hc}{\lambda_2} = w + (0.96 \times 10^{-19})$

$E_2 - E_1 = hc\left(\dfrac{1}{\lambda_2} - \dfrac{1}{\lambda_1}\right) = (0.64 \times 10^{-19})\text{J} - \text{sec}$

$h = \dfrac{\lambda_1 \times \lambda_2}{c(\lambda_1 - \lambda_2)} \times 0.64 \times 10^{-19} = \dfrac{(5000 \times 10^{-10})\,(4000 \times 10^{-10})}{3 \times 10^8\,(1000 \times 10^{-10})} \times 0.64 \times 10^{-19}\text{ J s}$

$= 4.27 \times 10^{-34}\text{ J} - \text{sec}$

6.(D)



7.(B) SO_4^{2-} : Central atom sp^3-hybridised hence tetrahedral.

$NF_3 : N - sp^3$-hybridised with one lone pair hence pyramidal.

$ICl_3 : I - sp^3d$-hybridised with two lone pair hence T-shaped.

$XeF_4 : Xe - sp^3d^2$-hybridised with two lone pairs hence square planar.

8.(C) $m_\alpha = 4m_p$; $q_\alpha = 2q_p$; $\lambda = \dfrac{h}{\sqrt{2\cdot m\cdot q\cdot V}}$

$$\lambda_\alpha = \frac{h}{\sqrt{2\times m_\alpha \times q_\alpha \times V}} = \frac{h}{\sqrt{2\times 4m_p \times 2q_p \times V}}$$

$$\lambda_p = \frac{h}{\sqrt{2\times m_p \times q_p \times V}} \quad \Rightarrow \quad \frac{\lambda_\alpha}{\lambda_p} = \sqrt{\frac{1}{4}\times\frac{1}{2}} = \frac{1}{2\sqrt{2}}$$

9.(A) As the polarizing power of cation decreasing, ionic character increases.

10.(B) There is a large difference between E_3 and E_4 values. Hence, number of valence electrons is 3.

11.(C) n = 3 ; n – l – 1 = 1 $\quad \Rightarrow \quad l = 1;\ L = \sqrt{l(l+1)}\cdot\dfrac{h}{2\pi}$

12.(B) For He^+ $(\infty \to 2)$

$$\frac{1}{\lambda} = \frac{1}{x} = R\times Z^2\left[\frac{1}{2^2} - \frac{1}{\infty^2}\right] \quad \Rightarrow \quad \frac{1}{x} = R$$

For $Li^{2+}(4\to 3)$

$$\frac{1}{\lambda} = R\times 3^2\left[\frac{1}{3^2} - \frac{1}{4^2}\right] = \frac{1}{x}\times 9\left[\frac{16-9}{9\times 16}\right] \quad \Rightarrow \quad \lambda = \frac{16x}{7}$$

13.(D) $\boxed{\uparrow}\quad \boxed{\uparrow \mid \downarrow\downarrow \mid \quad}$

14.(D) $a = 0, b = 5, a + b = 5$

15.(D) $\Delta x \cdot \Delta p = \dfrac{h}{4\pi} \quad \Rightarrow \quad \Delta v = \dfrac{h}{4\pi\, \Delta x \cdot m} = \dfrac{6.63\times 10^{-34}}{4\times 3.14\times 10^{-14}\times 9.1\times 10^{-31}} = 5.8\times 10^{9}$

16.(C) Radius (x_1) of n^{th} orbit for H-atom $= r_1 \times n^2$

If n is lower orbit, the next orbit = n + 1

and $x_2 = r_1(n+1)^2 = r_1(n^2 + 2n + 1)$

$x_2 - x_1 = r_1\,(2n+1)$

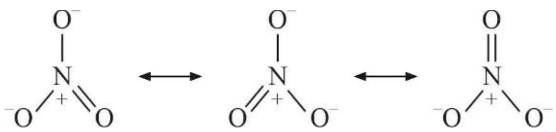
If n is higher orbit, then consecutive lower orbit = n – 1

and $x_2' = r_1(n-1)^2 = r_1(n^2 - 2n + 1)$

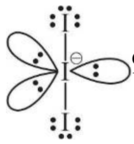
$x_1 - x_2' = r_1 \times n^2 - r_1\,(n^2 - 2n + 1) = r_1 \times (2n - 1)$

17.(A) In isoelectronic ions, the atomic size decreases as z/e ratio increases

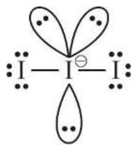
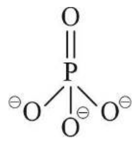
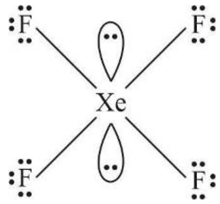
Ions	z/e ratio
S^{2-}	16/18
Cl^-	17/18
K^+	19/18
Ca^{2+}	20/18

18.(C) 

19.(B) The organic compound contains $16\sigma, 4\pi$ bonds**.**

20.(C)  9 lone pairs.

SECTION-2

1.(33)  x = 18  y = 1  z = 14

2.(4) SOF_4 is sp^3d hybrid but contains no lone pair

SeF_3^+ and $SeOCl_2$ contains lone pairs in sp^3 orbitals (no d-character)

XeF_5^+ is square pyramidal with one lone pair

BrF_6^- sp^3 d^3 hybridised is distorted octahedral containing one lone pair

TeF_5^- and ICl_4^- are sp^3d^2 hybrid with one and two lone pairs respectively.

3.(1) Total energy needed to convert one Mg atom into Mg^{2+}

(gas) ion $= IE_I + IE_{II}$

$= 22.681$ eV

$= 2268.1\,\text{kj mol}^{-1}$

12 mg of Mg $= 0.5 \times 10^{-3}$ mole

Amount of energy needed $= 2268.1 \times 0.5 \times 10^{-3} = 1.13405 \simeq 1$ $\approx 1\,\text{kJ}$

4.(6)

$$\left.\begin{array}{ccc} 10 & \rightarrow & 4 \\ 9 & \rightarrow & 4 \\ 8 & \rightarrow & 4 \\ 7 & \rightarrow & 4 \\ 6 & \rightarrow & 4 \\ 5 & \rightarrow & 4 \end{array}\right\} \text{Total 6 spectral lines in bracket series}$$

5.(48) 7s 5f 6d 7p $\rightarrow$ subshells to be filled in 7th period

Number of orbitals = 1 + 7 + 5 + 3 = 16

Number of electrons = 16 × 3 = 48

MATHEMATICS

SECTION-1

1.(B) $z_1^7 + z_2^7 + z_3^7 + ...z_7^7 = -1$

2.(D) $i^{1!} + i^{2!} + i^{3!} + i^{4!} + i^{5!} + ...i^{(101)!}$

$i^1 + i^2 + i^6 + \underbrace{1+1}_{98}$

$= 96 + i$

3.(C) $\underbrace{10 + 10^2 + 10^3 + ...}_{n} + \underbrace{2 + 4 + 6 + ...}_{n}$

$\frac{10(10^n - 1)}{9} + n^2 + n$

4.(C) $a = \omega, b = \omega^2$

5.(C) $p_r = \frac{r}{2}[2(r+1) + (r-1)(2r+2)]$

$p_r = r(r+1+(r-1)(r+1)$

$p_r = r(r + r^2)$

$p_r = r^2 + r^3$

$$\sum_{r=1}^{n} p_r = \frac{n(n+1)}{2}\left[\frac{2n+1}{3} + \frac{n(n+1)}{2}\right] = n(n+1)\left(\frac{4n+2+3n^2+3n}{12}\right) = \frac{n(n+1)(3n^2+7n+2)}{12}$$

$= \frac{n(n+1)(3n+1)(n+2)}{12}$

6.(D) General term = $-(4k-3)^2 - (4k-2)^2 + (4k-1)^2 + (4k)^2 = 4(8k-3)$ (if 4 terms are taken as terms)

$S_{20} = 32 \times \frac{5\times 6}{2} = 60(8-1)$

$S_{20} = 60 \times 7 = 420$

$S_{20} - 400 \Rightarrow S_{19} = 20$

7.(A) $\frac{(\bar{z})^2}{|z|^2} = \frac{(x-iy)^2}{x^2+y^2} = \frac{x^2 - y^2 - 2ixy}{x^2+y^2}$ First quadrant

8.(D) $S_n = n(n+1)(n+2)(n+3)$

$S_{n-1} = (n-1)(n)(n+1)(n+2)$

$T_n = 4n(n+1)(n+2)$

$\frac{1}{T_n} = \frac{1}{8}\left[\frac{1}{n(n+1)} - \frac{1}{(n+1)(n+2)}\right]$

$S_\infty = \frac{1}{16}$

9.(B) $\cos 2\alpha + i\sin 2\alpha - \cos\alpha + i\sin\alpha$

$\cos 2\alpha - \cos\alpha + i(\sin 2\alpha + \sin\alpha)$

$$-\left(2\sin\frac{3\alpha}{2}\sin\frac{\alpha}{2}\right) + i\,2\cos\frac{\alpha}{2}\sin\frac{3\alpha}{2} = i2\sin\frac{3\alpha}{2}\left(\cos\frac{\alpha}{2} + i\sin\frac{\alpha}{2}\right) = 2\sin\left(\frac{3\alpha}{2}\right)e^{i\left(\frac{\pi}{2}+\frac{\alpha}{2}\right)}$$

10.(C) $\frac{2z-n}{2z+n} = \frac{4i-1}{1} \Rightarrow \frac{4z}{2n} = \frac{4i}{2-4i}$

$$\frac{z}{n} = \frac{i(1+2i)}{(1-2i)(1+2i)}$$

$$\frac{z}{n} = -\frac{2+i}{5}$$

$$30 + iy = -\frac{2n}{5} + \frac{n}{5}i$$

$$-\frac{2n}{5} = 30$$

$$n = -75$$

11.(B) $S = t + 2t^2 + 3t^3 + ...9t^9$

$tS = t^2 + 2t^3 + ...8t^9 + 9t^{10}$

$S(1-t) = t + t^2 + t^3 + ...t^9 - 9t^{10}$ $\quad\quad \{t, t^2, t^3.....t^9$ is sum of 9^{th} root of unity$\}$

$S(1-t) = -9t^{10}$

$$S = -\frac{9t}{(1-t)}$$

$$|S| = \frac{9}{|1-t|} = \frac{9}{|1-\cos\theta - i\sin\theta|} = \frac{9}{2\sin 20^\circ}$$

12.(C) $\left|z + \frac{2}{z}\right| = 1 \quad\quad |z| = r$

$$\left||z| - \frac{2}{|z|}\right| \le \left|z + \frac{2}{z}\right|$$

$$\left|r - \frac{2}{r}\right| \le 1 \;\Rightarrow -1 \le r - \frac{2}{r} \le 1$$

$r \in [1, 2]$

$|z| \in [1, 2]$

13.(D) $x^{11} - 1 = (x-1)(x-\lambda_1)(x-\lambda_2)......(x-\lambda_{10}) \quad \Rightarrow \quad (\omega^2-\lambda_1)(\omega^2-\lambda_2).........(\omega^2-\lambda_{10}) = \dfrac{\omega^{22}-1}{\omega^2-1}$

$$(\omega-\lambda_1)(\omega-\lambda_2)............(\omega-\lambda_{10}) = \frac{\omega^{11}-1}{\omega-1}$$

$$\frac{(\omega^2-\lambda_1)(\omega^2-\lambda_2)..........(\omega^2-\lambda_{10})}{(\omega-\lambda_1)(\omega-\lambda_2)................(\omega-\lambda_{10})} = \frac{\frac{(\omega-1)}{(\omega^2-1)}}{\frac{\omega^2-1}{(\omega-1)}} = \frac{(\omega-1)(\omega-1)}{(\omega-1)(\omega+1)(\omega-1)(\omega+1)} = \frac{1}{\omega^4} = \frac{1}{\omega} = \omega^2$$

14.(A) $1 + 2^2 + 3 + 4^2 + 5 + 6^2......+(2k)^2 = \dfrac{2k(2k+4)(4k+1)}{12}$

$$1 + 2^2 + 3 + 4^2 + ...+(2k)^2 + 2k + 1 = \frac{2k(2k+4)(4k+1)}{12} + (2k+1)$$

$$\frac{(n-1)(n+3)(2n-1)}{12} + n$$

15.(C) $a + b + c = \dfrac{n(\lambda+N)}{2} \Rightarrow \dfrac{3(\lambda+N)}{2} = 30$

$\lambda + N = 20$

$$\frac{1}{p} + \frac{1}{q} + \frac{1}{r} = \frac{3}{2}\left(\frac{1}{\lambda} + \frac{1}{N}\right)$$

$$\frac{5}{6}=\frac{3}{2}\left(\frac{20}{\lambda N}\right)\Rightarrow \lambda N=36$$

16.(B) n^{th} row's 1^{st} term =

1, 2, 4, 7, 11 ……………

$T_n = an^2 + bn + c$

n^{th} rows last term $=\dfrac{n^2-n+2+n-1}{2}=\dfrac{n^2+n}{2}$

$$\frac{n^2-n+2}{2}\le 100\le \frac{n^2+n}{2} \Rightarrow n=14$$

17.(A) $z=\dfrac{2^8\left(\sqrt{3}+i\right)^8}{\left(\sqrt{2}\right)^6\left(e^i-\dfrac{6\pi}{4}\right)}+\dfrac{\left(\sqrt{2}\right)^6\left(e^i-\dfrac{6\pi}{4}\right)}{2^8\left(\sqrt{3}+i\right)^8}$

$$z=\frac{2^{16}e^i\left(\frac{8\pi}{6}+\frac{6\pi}{4}\right)}{2^3}+\frac{2^3e^i\left(\frac{6\pi}{4}+\frac{8\pi}{6}\right)}{2^{16}}$$

$$z=2^{13}\,e^{i\,5\pi/6}+\frac{1}{2^{13}}e^{i\;5\pi/6}$$

$$z=\left(2^{13}+\frac{1}{2^{13}}\right)e^{i\;5\pi/6}$$

18.(A) $T_n=\dfrac{1}{n\sqrt{n+1}+(n+1)\sqrt{n}}=\dfrac{(n+1)\sqrt{n}-(\sqrt{n+1})\sqrt{n}}{n(n+1)}=\dfrac{1}{\sqrt{n}}-\dfrac{1}{\sqrt{n+1}}$

$$S=\frac{1}{3}-\frac{1}{24}=\frac{7}{24}$$

19.(B) $S_{n+2}-S_n=(n+2)^2$

$S_{10}-S_8=10^2$

$S_8-S_6=8^2$

$S_6-S_4=6^2$

$S_4-S_2=4^2$

$S_{10}-S_2=4^2+6^2+8^2+10^2$

$\Rightarrow \quad S_{10}=4(1^2+2^2+3^2+4^2+5^2)$

20.(A) $z^6+z^3+1=0$

$z^3=\omega$ or $z^3=\omega^2$

$z^3=e^{i\left(\frac{2\pi}{3}+2k\pi\right)}$ or $z^3=e^{i\left(\frac{4\pi}{3}+2k\pi\right)}$

$z=e^{i\frac{2\pi}{9}}, e^{i\frac{8\pi}{9}}, e^{i\frac{14\pi}{9}}, e^{i\frac{4\pi}{9}}, e^{i\frac{10\pi}{9}}, e^{i\frac{16\pi}{9}} \quad\Rightarrow\quad \theta=\dfrac{10\pi}{9}$

SECTION-2

1.(3) $a = p - 2$

$ar^2 = p + 6$

$\frac{p-2}{p+6} = \frac{1}{r^2}$

$\frac{p-2}{p+6} = 9 \text{ or } \frac{1}{9}$

$\frac{\frac{a+ar^2}{2}}{ar} = \frac{5}{3}$

$\frac{r^2+1}{2r} = \frac{5}{3}$

$3r^2 + 3 = 10r$

$r = 3,\ r = \frac{1}{3}$

2.(6) $1 - 4 + 4i - a - 2ia + ib = 0$

$-3 - a + i(4 - 2a + b) = 0$

$a = -3 \qquad b = -10$

$(1+2i)\lambda = -10i \quad \Rightarrow \quad \lambda = \frac{-10i(1-2i)}{(1+2i)(1-2i)}$

$\lambda = -4 - 2i$

3.(27) $\left|a + b\omega + c\omega^2\right|^2 = \frac{(a-b)^2 + (b-c)^2 + (c-a)^2}{2}$

$a = 3n,\ b = 3n+3,\ c = 3n+6$

$\left|a + b\omega + c\omega^2\right|^2 = 27$

4.(3) $\frac{\frac{a}{2} + \frac{b}{3} + \frac{c}{4} + \frac{d}{9} = \frac{2}{3}}{4} \geq \sqrt[4]{\frac{abcd}{2 \cdot 3 \cdot 4 \cdot 9}}$

AM = GM

$\frac{a}{2} = \frac{b}{3} = \frac{c}{4} = \frac{d}{9} = \frac{2}{3 \times 4}$

$a + b + c + d = 3$

5.(15) $\frac{1}{8}\left(\frac{1}{1^3} + \frac{1}{2^3} + \frac{1}{3^3}........\right) + \frac{1}{1^3} + \frac{1}{3^3} + \frac{1}{5^3}..... = p$

$\frac{p}{8} + q = p$

$q = \frac{7}{8}p \;\Rightarrow\; \frac{p}{q} = \frac{8}{7}$